



DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF TECHNOLOGY
THIRD SESSIONAL
SUBJECT: (CE-611) COMPUTER NETWORKS

Examination : B.Tech Semester VI
Date : 21/03/2023
Time : 2:30 PM – 3:45 PM

Seat No. : CE137
Day : Tuesday
Max. Marks : 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

Q.1 Do as directed.

- CO3 A (a) One hundred stations on pure ALOHA network share a 1-Mbps channel. If frames are 1000 bits long, find the throughput, if each station is sending 10 frames per second. [12]
13-5
- CO4 E (b) An Ethernet MAC sub-layer receives 42 bytes of data from upper layer. Is padding needed in the Ethernet frame, and if so, how many bytes? [2]
A
- CO5 N (c) "Ethernet has imposed restrictions on both the minimum and maximum lengths of a frame." Justify. [2]
to be
- CO1 E (d) If a channel is absolutely error free, do we need sequence numbers and acknowledgments for Segments? Justify your answer. [2]
no
- CO3 U (e) What are UA and MTA? What do they do? [2]
- CO1 A (f) $rtt=40$, measured $rtt=38$, $rttvar=1$. What will be value of RTO if Jacobson's calculations are taken into account? [2]
44-75

Q.2 Attempt Any TWO from the following questions.

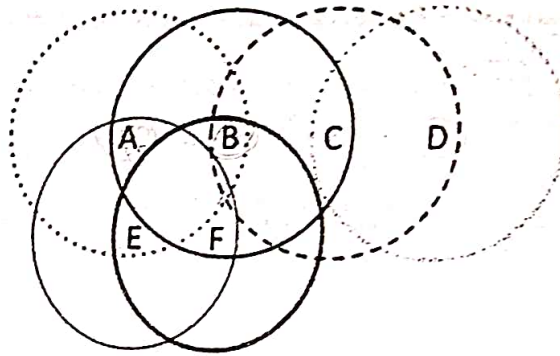
- CO4 N (a) Consider a network connecting two systems located 12000 kilometers apart. The bandwidth of the network is 500×10^6 bits per second. The propagation speed of the media is 4×10^6 meters per second. It is needed to design a Go-Back-N sliding window protocol for this network. The average frame size is 10^7 bits. The network is to be used to its full capacity. Assume that processing delays at nodes are negligible. How many minimum bits are required in the sequence number field? [12]
- CO5 U (b) Running TCP across wireless networks may lead to performance problems. Why? Discuss two possible solutions for the problem with their limitations if there is any. [6]
bs, bw
- CO3 R (c) Answer with regard to TCP. [6]
pwr, dlec, timer, ack, fin, sink, for 2, (1st) tw
1. What is usefulness of time waited timer?
 2. What is congestion avoidance?
 3. What will be triggering event and next state when TCP is in FIN WAIT 1 state?

Q.3 Answer the following questions.

- CO2 A (a) Discuss binary countdown protocol. Explain Mok and Ward version of binary countdown. Consider that a LAN uses Mok and Ward's version of binary countdown. At a certain instant, the ten stations have the virtual station numbers 8, 2, 4, 5, 1, 7, 3, 6, 9, and 0. The next three stations to send are 4, 3, and 9, in that order. What are the new virtual station numbers after all three have finished their transmissions? [12]
8 2 5 1 7 6 0 4 3 9
- CO3 N (b) Explain the role of SMI, MIB and SNMP in network management. [6]

CO1 N (c) Consider the following wireless topology, consisting of 6 nodes. Each node's transmission range is indicated by circles around it. [3]

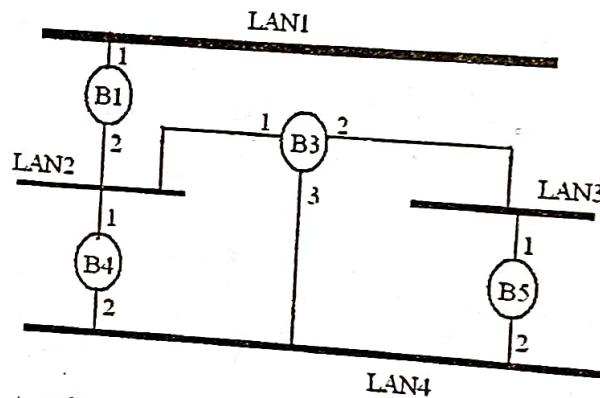
1. When node A transmits to node B, list the potential hidden terminals from A and exposed terminals. *C, F, D, E*
2. When node B transmits to node C, list the potential hidden terminals from B and exposed terminals. *E, F, A*



OR

Q.3 Answer the following questions.

- CO2 A (a) Discuss the problem created by two parallel transparent bridges in the topology. Find the spanning tree for the system shown in figure below. Choose B1 as Root Bridge. Show the method. Identify forwarding and blocking ports of the bridges. [12]



- CO3 N (b) Explain 1-persistent and p-persistent CSMA. [3]
- CO1 N (c) Suppose measurements made on a slotted ALOHA channel for a very large number of users shows that on average 20% of the slots are idle. [3]
1. What is the channel load G?
 2. What is the throughput of the channel?
 3. Is this channel overloaded or under loaded?